

Structural Companion to Algebraic Exponential Integrals: Audit, Detailed Lemmas, and Directions for Generalization

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Abstract

This companion records the material removed from the streamlined base paper *Algebraic Exponential Integrals: Transcendence and Linear Independence*. It has three purposes. First, it audits the earlier 1,832-line draft and records the parameter absorptions and proof compression. Second, it preserves the stronger structural results that are not needed for the shortest base proof: the full Brieskorn lattice, a detailed Fourier–Laplace semisimplicity argument, the complete horizontal endomorphism algebra, multiplicity-free decomposition, the relation differential submodule of relations and its constant projector, and the exact Cauchy-transform criterion. Third, it separates polynomial-specific calculations from mechanisms likely to survive for finite maps of curves, larger families of exponential period functions, and algorithmic E -function problems.

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1 Vocabulary and scope

The revised base paper follows the terminology used in the closest E -function and exponential-period literature. A parameter integral $\int_{\gamma} e^{-zf} \omega$ is an *exponential period function*; f is a regular function and, in the one-variable case considered here, a polynomial [6, 4]. The geometric language is that of exponential twists, twisted de Rham cohomology, Fourier–Laplace transforms, and topological or relative chains [8, 12]. For the final analytic step, the established terms are moments, the polynomial moment problem, Cauchy transforms, zero-cycles, inverse branches, and polynomial constellations [10, 7].

Accordingly, this companion uses *polynomial q* , *values of q* , *critical values*, *zero-cycle*, and *moment identities*. The word *phase* is reserved for genuine stationary-phase or oscillatory-asymptotic arguments and is not used for the basic algebraic data here. The standard word *constellation* is used only for the preimage of a star joining a regular value to the critical values. When extra arms are added to reach $q(\text{supp } \Delta)$, we refer simply to the enlarged tree T and its lifted graph $q^{-1}(T)$.

2 Audit of the earlier draft

The audited source is the August 2026 manuscript `algebraic_exponential_integrals.tex`, containing 1,832 source lines. The revised base source contains about 710 lines, including its expanded bibliography, and compiles to nine pages in the present format.

2.1 Verdict

No fatal gap was found in the main proof chain

$$\begin{aligned} &\text{Beukers lifting} \implies \text{constant splitting} \implies \text{exponential polynomial,} \\ &\text{exponential polynomial} \implies \text{weighted moments} \implies \text{Cauchy-transform vanishing criterion.} \end{aligned}$$

The highest-risk points remain the standard but convention-sensitive Fourier–Laplace identification, the localization of simple Weyl modules, and the lateral-jump argument on a ramified lifted graph. They are recorded in expanded form below so that they can be audited independently.

The principal problem in the earlier draft was not a false theorem but a mismatch between two goals: it used the foundational theorem as the vehicle for developing a much larger structural toolkit. The base proof was therefore longer than necessary, while useful generalization mechanisms were buried inside a theorem that did not require their full strength.

2.2 Parameter absorptions and genuine simplifications

Item	Earlier treatment	Streamlined treatment
Evaluation parameter ξ	Carried through the main theorem and the proof.	Replace q by ξq and prove only the value at $z = 1$. Since $I_{\xi q, \Delta}(z) = I_{q, \Delta}(\xi z)$, no information is lost.
Affine normalization of q	Translate to remove the x^{n-1} term and scale the leading coefficient to $1/n$.	Unnecessary. For arbitrary leading coefficient, Euclidean division still gives $h_j = x^{j+1}/n + O(x^j)$, hence the same residue spectrum.
Normalization of the inhomogeneous term	Define scaled vectors $n(h_0, \dots, h_{n-2})$ and use the prefactor $1/(nz)$.	Define $\mathbf{h} = (h_j)$ directly and write $z^{-1} \sum \mathbf{b}_\lambda e^{\lambda z}$. The factor n has no mathematical role.
Special identity $h_0 = x/n$	Used to motivate an unnecessary scaling of the inhomogeneous term.	Not needed anywhere in the rigidity argument.
General Beukers statement	State the full homogeneous-polynomial lifting theorem.	Only the linear lifting statement at $z = 1$ is used.
Horizontal endomorphisms	Compute the complete algebra and deduce multiplicity-free semisimplicity.	The base theorem needs only that a horizontal projector is constant. The stronger theorem is retained here.
Cauchy geometry	Prove an exact Cauchy-transform characterization of all functional relations.	The base theorem needs only the implication from weighted moments to functional vanishing. The equivalence is retained here.
Height bookkeeping	A separate joint-height lemma and a fully quantified denominator construction.	The base paper uses the exact coefficient formula plus a one-paragraph exponential denominator estimate. The full certificate is retained in Section 4.

2.3 Corrections of emphasis and notation

1. The base theorem should be advertised as an exact *transfer* of value relations to functional relations. An explicit closed-form classification of every functional relation is a separate geometric problem. The distinct-regular-value criterion gives an important sufficient case, not a general combinatorial classification.
2. The symbols used for affine scaling in the old normalization argument collided with the support-point symbol α . Removing the normalization also removes this avoidable notational hazard.
3. The semisimplicity proof is correct only after the reduced constant summand is tracked through Fourier transform: the constant connection becomes a module supported at the

Fourier origin and vanishes after generic localization. This point should be stated, not left implicit.

4. Generic localization of a simple Weyl module must be justified. Exactness alone does not say that a nonzero localization remains simple. The needed one-line argument is included in Proposition 5.1.
5. The Cauchy-transform proof requires integrability at ramified vertices. The estimate $x'(t) = O(|t - c|^{1/r-1})$ is load-bearing and should remain visible in any compressed version.

3 Intrinsic moment module without normalization

Let $q \in \overline{\mathbb{Q}}[x]$ have degree $n \geq 2$, put $K = \overline{\mathbb{Q}}(z)$, and define

$$\mathcal{M}_q = K[x]/(\partial_x + zq'(x))K[x]. \quad (3.1)$$

The quotient is a K -vector space, not a quotient ring. Write $[p]$ for the class of p .

Lemma 3.1 (Brieskorn basis). *The classes $[1], [x], \dots, [x^{n-2}]$ form a K -basis of $\mathcal{M}_q$. Moreover,*

$$\mathbb{C}[z, z^{-1}, x]/(\partial_x + zq')\mathbb{C}[z, z^{-1}, x]$$

is free over $\mathbb{C}[z, z^{-1}]$ with the same basis. Differentiation in z defines the connection

$$\nabla_{\partial_z}[p] = [\partial_z p + qp].$$

Proof. The identity $[zq'p] = -[p']$ reduces every monomial of degree at least $n - 1$. If $p \neq 0$, then $(\partial_x + zq')p$ has degree $\deg p + n - 1$, so its leading term cannot cancel. Hence the image contains no nonzero polynomial of degree below $n - 1$. Finally, $[\partial_z + q, \partial_x + zq'] = 0$, so the connection is well defined. $\square$

For $0 \leq j \leq n - 2$, divide

$$qx^j = h_j q' + r_j, \quad \deg r_j < n - 1, \quad (3.2)$$

and define

$$C(x^j) = r_j, \quad D(x^j) = -h'_j.$$

Then C is multiplication by q on the Jacobian algebra $\mathbb{C}[x]/(q')$, and the connection matrix is $C + D/z$.

Lemma 3.2 (Unnormalized residue spectrum). *For every polynomial q of degree n , without any translation or scaling, D is triangular in the monomial basis with diagonal*

$$-\frac{1}{n}, -\frac{2}{n}, \dots, -\frac{n-1}{n}.$$

Proof. If $q(x) = cx^n + O(x^{n-1})$, then $q'(x) = ncx^{n-1} + O(x^{n-2})$. Consequently, $h_j = x^{j+1}/n + O(x^j)$ and $-h'_j = -(j+1)x^j/n + O(x^{j-1})$. $\square$

4 Full arithmetic certificate for the moment functions

Fix a finite degree-zero zero-cycle $\Delta = \sum_{\alpha \in S} d_\alpha [\alpha]$ with coefficients in $\overline{\mathbb{Q}}$, and choose a finite piecewise C^1 one-chain Γ with $\partial\Gamma = \Delta$. Put

$$U_j(z) = \int_{\Gamma} x^j e^{zq(x)} dx = \sum_{m \geq 0} u_{j,m} \frac{z^m}{m!}.$$

Lemma 4.1 (Joint height from one denominator). *Let K_0 be a number field and let $\alpha_0, \dots, \alpha_m \in K_0$. If $\delta\alpha_k \in \mathcal{O}_{K_0}$ for every k , then*

$$h(\alpha_0, \dots, \alpha_m) \leq \log \delta + \log \max \left(1, \max_{k, \sigma} |\sigma(\alpha_k)| \right).$$

Proof. Use the projective coordinates $(1 : \alpha_0 : \dots : \alpha_m) = (\delta : \delta\alpha_0 : \dots : \delta\alpha_m)$. All finite-place contributions are at most 1, and the infinite-place contribution is bounded by the displayed quantity. $\square$

Proposition 4.2 (Explicit E -function certificate). *Each U_j is an E -function. More precisely, there is a constant κ_j such that*

$$h(u_{j,0}, \dots, u_{j,m}) \leq \kappa_j(m+1).$$

Proof. Write

$$q(x)^m = \sum_{k=0}^{nm} \gamma_{m,k} x^k.$$

Since a primitive of x^{j+k} is $x^{j+k+1}/(j+k+1)$,

$$u_{j,m} = \sum_{k=0}^{nm} \gamma_{m,k} \frac{\sum_{\alpha \in S} d_\alpha \alpha^{j+k+1}}{j+k+1}. \quad (4.1)$$

Choose a number field K_0 containing all data and an integer $d \geq 1$ such that the coefficients of dq , all $d\alpha$, and all dd_α are integral. Then

$$\delta_m = d^{(n+1)m+j+2} \text{lcm}(1, \dots, nm+j+1)$$

is a common denominator for $u_{j,0}, \dots, u_{j,m}$. Indeed, $d^m \gamma_{m,k}$ is integral, and the spare factor d^{nm-k} makes the exponent uniform in (4.1).

For archimedean embeddings, choose a disk containing every conjugate of every support point. Integrating the conjugate polynomial along the straight segment from 0 to the support point gives $|\sigma(u_{j,m})| \leq C^{m+j+2}$. Since $\log \text{lcm}(1, \dots, N) = O(N)$, Lemma 4.1 gives the height bound.

The integration-by-parts system

$$\mathbf{U}' = \left(C + \frac{D}{z} \right) \mathbf{U} + \frac{1}{z} \sum_{\lambda} \mathbf{b}_{\lambda} e^{\lambda z}$$

shows that the $\overline{\mathbb{Q}}(z)$ -span of the moments together with the exponentials $e^{\lambda z}$ for $\lambda \in q(\text{supp } \Delta)$ is stable under differentiation. Hence each U_j is holonomic, completing the E -function certificate. $\square$

5 Expanded Fourier–Laplace semisimplicity

Let Σ be the critical-value set, $V_0 = \mathbb{A}_t^1 \setminus \Sigma$, and let $\mathcal{L}$ be the permutation local system of the unramified cover $q^{-1}(V_0) \rightarrow V_0$. Let $\mathcal{L}^0 = \ker(\text{Tr} : \mathcal{L} \rightarrow \mathbb{C}_{V_0})$.

Proposition 5.1 (Expanded semisimplicity proof). *The differential module over $\mathbb{C}(z)$ with matrix $C + D/z$ is semisimple.*

Proof. The cover is connected, so its monodromy acts transitively on a fiber. The permutation representation has finite image and splits as

$$\mathcal{L} \simeq \mathbb{C}_{V_0} \oplus \mathcal{L}^0.$$

Maschke’s theorem makes $\mathcal{L}^0$ semisimple. Since q is finite, it is proper and small. Therefore

$$Rq_* \mathbb{C}_{\mathbb{A}_x^1}[1] \simeq \text{IC}(\mathcal{L}),$$

and the decomposition of $\mathcal{L}$ into irreducibles gives a decomposition of $q_* \mathcal{O}_{\mathbb{A}_x^1}$ into simple regular holonomic modules under the Riemann–Hilbert correspondence.

Use the Fourier convention $t \mapsto -\partial_\tau$, $\partial_t \mapsto \tau$. Fourier transform is an exact autoequivalence of Weyl modules and preserves simple objects. The constant connection transforms to a module supported at $\tau = 0$ and hence disappears after localization to $\mathbb{C}(\tau)$.

We record the localization point. Let M be a simple Weyl module and $S = \mathbb{C}[\tau] \setminus \{0\}$. The S -torsion of M is a Weyl submodule: if $p(\tau)m = 0$, then $p(\tau)^2 \partial_\tau m = 0$. Hence either $S^{-1}M = 0$ or M injects into $S^{-1}M$. In the latter case, every nonzero differential submodule of $S^{-1}M$ meets M after clearing a denominator; the intersection is a nonzero Weyl submodule and therefore equals M . Thus every nonzero localized simple summand remains irreducible.

To identify the localized transform, factor q as its graph embedding followed by projection. The graph generator δ_q satisfies

$$(t - q(x))\delta_q = 0, \quad (\partial_x + q'(x)\partial_t)\delta_q = 0.$$

After Fourier transform it is the exponential generator $e^{-\tau q(x)}$. On coefficient polynomials, the relative differential and connection are

$$\partial_x - \tau q'(x), \quad \partial_\tau - q(x).$$

The relative de Rham complex has no negative-degree cohomology because $-\tau q'p$ has larger x -degree than p' for $p \neq 0$. Its degree-zero cohomology is

$$\mathbb{C}(\tau)[x]/(\partial_x - \tau q')\mathbb{C}(\tau)[x].$$

Pulling back by $\tau = -z$ gives (3.1) with connection $\partial_z + q$. By Lemma 3.1, its matrix is $C + D/z$. It is therefore a direct sum of irreducible differential modules. $\square$

6 The complete horizontal endomorphism algebra

For a critical point τ , put $m_\tau = \text{ord}_\tau q'$ and $\lambda_\tau = q(\tau)$, and define

$$g_{\tau,k}(x) = \frac{q'(x)}{(x - \tau)^k}, \quad 1 \leq k \leq m_\tau.$$

Lemma 6.1 (Critical basis and compressed residue). *The $g_{\tau,k}$ form a basis of $\mathcal{V} = \mathbb{C}[x]_{<n-1}$. They diagonalize $\mathbb{C}$:*

$$\mathbb{C}g_{\tau,k} = \lambda_{\tau}g_{\tau,k}.$$

For a fixed critical value λ , the compression of $\mathbb{D}$ to the λ -eigenspace preserves each critical-point summand. On the τ -summand its triangular diagonal is

$$-\frac{m_{\tau}}{m_{\tau}+1}, -\frac{m_{\tau}-1}{m_{\tau}+1}, \dots, -\frac{1}{m_{\tau}+1}.$$

Proof. The Chinese remainder theorem gives

$$\mathbb{C}[x]/(q') \simeq \bigoplus_{\tau} \mathbb{C}[x]/(x-\tau)^{m_{\tau}}.$$

For fixed τ , the images of $g_{\tau,k}$ vanish in all other factors and form a triangular basis in the τ -factor. Since $q - \lambda_{\tau}$ vanishes to order $m_{\tau} + 1$,

$$qg_{\tau,k} = \frac{q - \lambda_{\tau}}{(x - \tau)^k} q' + \lambda_{\tau}g_{\tau,k},$$

which proves the eigenvalue identity.

Write $y = x - \tau$ and $q - \lambda_{\tau} = ay^{m_{\tau}+1} + O(y^{m_{\tau}+2})$. Then

$$g_{\tau,k} = a(m_{\tau} + 1)y^{m_{\tau}-k} + O(y^{m_{\tau}-k+1}),$$

while

$$-\left(\frac{q - \lambda_{\tau}}{(x - \tau)^k}\right)' = -a(m_{\tau} + 1 - k)y^{m_{\tau}-k} + O(y^{m_{\tau}-k+1}).$$

This gives the diagonal. If $\sigma \neq \tau$ has the same critical value, the displayed quotient vanishes at σ to order $m_{\sigma} + 1$; its derivative therefore vanishes modulo $(x - \sigma)^{m_{\sigma}}$, excluding cross terms. $\square$

Theorem 6.2 (Horizontal-endomorphism rigidity). *If $P \in M_{n-1}(\mathbb{C}(z))$ satisfies*

$$P' = \left[C + \frac{D}{z}, P \right],$$

then P is constant. Hence the horizontal endomorphism algebra is

$$Z(C) \cap Z(D).$$

The same assertion holds for the dual module.

Proof. Finite nonzero poles are excluded by pole order. At zero, a pole of order m gives $-mP_{-m} = [D, P_{-m}]$, impossible because the spectrum of $\text{ad } D$ lies in $(-1, 1)$. Thus P is polynomial.

Transpose and write $Q = Q_d z^d + Q_{d-1} z^{d-1} + \dots$. If $d \geq 1$, the top two coefficient equations are

$$[Q_d, C] = 0, \quad dQ_d = [Q_{d-1}, C] + [Q_d, D].$$

Projecting to a nonzero critical-value block of Q_d and using Lemma 6.1 yields

$$dQ_{d,\lambda} = [Q_{d,\lambda}, \overline{D}_{\lambda}].$$

Every eigenvalue of the commutator on the right is a difference of two numbers in $(-1, 0)$ and therefore lies in $(-1, 1)$. It cannot equal the positive integer d . Hence $d = 0$. Substitution gives $[C, P] = [D, P] = 0$. $\square$

Corollary 6.3 (Multiplicity-free semisimplicity). *The horizontal endomorphism algebra is commutative. Consequently, the semisimple moment connection is multiplicity-free.*

Proof. The matrix D has distinct eigenvalues, so its centralizer is commutative. A repeated simple constituent of a semisimple differential module would contribute a noncommutative full matrix algebra to the endomorphism algebra. $\square$

7 The relation submodule and its constant projector

Let $\mathbf{U}$ satisfy the inhomogeneous moment system and put

$$\mathcal{E} = \text{span}_{\mathbb{C}(z)}\{e^{\lambda z} : \lambda \in \Lambda\}.$$

Define

$$\mathcal{R} = \{\phi \in \mathbb{C}(z)^{1 \times (n-1)} : \phi \mathbf{U} \in \mathcal{E}\}.$$

Lemma 7.1 (Differential submodule of relations). *The space $\mathcal{R}$ is a differential submodule of the dual homogeneous connection.*

Proof. Write $A = C + D/z$ and $\mathbf{f} = z^{-1} \sum_{\lambda} \mathbf{b}_{\lambda} e^{\lambda z}$. Then

$$(\phi \mathbf{U})' = (\phi' + \phi A) \mathbf{U} + \phi \mathbf{f}.$$

The left side and the last term belong to $\mathcal{E}$ whenever $\phi \in \mathcal{R}$. $\square$

Proposition 7.2 (Constant projector mechanism). *The relation module has the form*

$$\mathcal{R} = W \otimes_{\mathbb{C}} \mathbb{C}(z)$$

for a constant subspace W invariant under C and D .

Proof. By Proposition 5.1, the dual moment module is semisimple, so $\mathcal{R}$ has a horizontal complement and a horizontal idempotent projector. By Theorem 6.2, the projector is constant. Its image is the required W . $\square$

Remark 7.3 (Why this mechanism should be preserved). The base theorem uses the projector only once, to pass from a lifted row whose value at 1 is $\mathbf{e}_0$ to the constant row $\mathbf{e}_0$ itself. In larger systems of exponential period functions, however, the same projector identifies the full differential submodule of functional relations. It is therefore a natural interface with uniform relation algorithms and with decompositions into summands detected or annihilated by the selected period functions.

8 Exact Cauchy-transform criterion

Let Δ be a finite degree-zero zero-cycle with complex coefficients, let Γ be a finite piecewise C^1 one-chain with $\partial \Gamma = \Delta$, and choose a finite tree T containing all critical values of q and all values $q(\alpha)$ with $\alpha \in \text{supp } \Delta$. Choose a lifted graph chain $\tilde{\Gamma} \subset q^{-1}(T)$ with the same boundary. On an oriented open edge e , write $\tilde{\Gamma}$ as a linear combination of inverse branches $x_{e,i}(t)$ and put

$$\psi_e(t) = \sum_i f_{e,i} x'_{e,i}(t) = \sum_i \frac{f_{e,i}}{q'(x_{e,i}(t))}.$$

Thus $\psi_e(t) dt$ is the density contributed by the lifts of e .

Theorem 8.1 (Exact Cauchy-transform criterion). *For fixed T and $\tilde{\Gamma}$,*

$$I_{q,\Delta}(z) \equiv 0 \iff \psi_e \equiv 0 \text{ on every open edge } e.$$

More generally, if $0 \neq H \in \mathbb{C}[t]$ and

$$\int_{\Gamma} q(x)^m H(q(x)) dx = 0 \quad (m \geq 0),$$

then every ψ_e vanishes.

Proof. For $s \notin T$, define

$$\mathcal{C}_H(s) = \int_{\tilde{\Gamma}} \frac{H(q(x))}{s - q(x)} dx.$$

The moment identities make its Laurent expansion at infinity zero. Since $\mathbb{C} \setminus T$ is connected, $\mathcal{C}_H$ vanishes there. Edgewise change of variables gives

$$\mathcal{C}_H(s) = \sum_e \int_e \frac{H(t)\psi_e(t)}{s - t} dt.$$

At an interior point of one edge, contributions from all other edges extend holomorphically across a small disk. The local Cauchy integral has lateral jump $2\pi i \varepsilon_e H(t)\psi_e(t)$. Both lateral values vanish, so $H(t)\psi_e(t) = 0$. Away from the finite zero set of H , this gives $\psi_e = 0$, and analyticity extends the conclusion to the entire edge. The exponential integral is the edgewise Laplace transform of the densities, proving one direction. The other is immediate. Taking $H = 1$ after $I_{q,\Delta} \equiv 0$ gives the converse implication in the equivalence. This is the standard Cauchy-transform mechanism used in the polynomial moment literature; compare [7, 11]. $\square$

Corollary 8.2 (Functional independence over distinct regular values). *If the support points $a_0, \dots, a_r$ have pairwise distinct regular values under q , then the zero-cycle $\sum_i d_i [a_i]$ has identically zero exponential integral only when all $d_i = 0$.*

Proof. Give each $q(a_i)$ a separate terminal arm. On that arm, the only nonzero terminal coefficient is d_i , and the density equals $d_i/q'(x_i(t))$. Apply Theorem 8.1. $\square$

9 What is likely to generalize

The following division should guide later papers on exponential period functions.

9.1 Robust structural mechanisms

1. **Differential submodule of relations and projector.** This is not tied to a scalar integral. The same construction applies to an inhomogeneous differential system whose homogeneous module is semisimple and whose inhomogeneous term lies in a finite-dimensional differential submodule generated by exponential functions.
2. **Nonresonant residue criterion.** The pole arguments use only that residue eigenvalue differences avoid nonzero integers, locally at finite points and after compression at infinity. This suggests an abstract nonresonant constant-splitting criterion.

3. **Fourier–Laplace semisimplicity.** Finite monodromy before Fourier transform is a particularly transparent source of semisimplicity. More general semisimple local systems on curves should produce analogous transformed modules.
4. **Cauchy jump detection.** Moment vanishing is converted into vanishing of a Cauchy transform, and local jumps recover the density. The argument should extend to algebraic finite covers of curves once a suitable cut graph and local branches are available.

9.2 Polynomial-specific pieces

1. The monomial Brieskorn basis $1, x, \dots, x^{n-2}$ and the division algorithm by q' use the affine line essentially.
2. The critical basis $q'/(x - \tau)^k$ and the explicit fractions $-r/(m + 1)$ are special to one-variable polynomial ramification.
3. The chosen tree T lies in the target plane. On a positive-genus curve it must be replaced by a cut graph, and closed-cycle periods may remain after all edge densities vanish.

9.3 Algorithmic outputs worth preserving

For a concrete polynomial q , the following data are explicit and small:

$$C, \quad D, \quad \{(\tau, m_\tau, \lambda_\tau)\}, \quad \{\overline{D}_\lambda\}, \quad \{\mathbf{b}_\lambda\}.$$

They provide certificates for residue nonresonance, constant projectors, and detection of boundary coefficients without reconstructing a large minimal scalar operator. This is the natural point of contact with the Adamczewski–Rivoal and Bostan–Rivoal–Salvy exceptional-value algorithms.

AI-assisted research disclosure

This companion was developed through interactive work between Christopher D. Long and ChatGPT 5.6 Pro. ChatGPT assisted in proof compression, line-by-line auditing, symbolic stress tests, organization, and drafting. The AI system is not an author. The authors bear full responsibility for all mathematical claims and any remaining errors.

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